

FREDDI on Jupyter Notebook

Step-by-Step Execution Guide

FREDDI is an astrophysical accretion disk evolution code commonly used for studying time-dependent accretion disk physics in compact objects such as black holes and X-ray binaries. This guide explains how to install and execute FREDDI inside a Jupyter Notebook environment.

1. Install Miniforge / Conda

Open a Linux terminal and execute:

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)$(uname -m).sh"
bash Miniforge3-$(uname)$(uname -m).sh
conda init
```

2. Create a Python Environment

After restarting the terminal:

```
conda create -n freddi_env python=3.11 jupyter numpy scipy matplotlib astropy
conda activate freddi_env
```

3. Clone the FREDDI Repository

Download the code from GitHub:

```
git clone https://github.com/hombit/freddi.git
cd freddi
```

4. Install FREDDI

Install the package:

```
pip install .
```

5. Launch Jupyter Notebook

Start Jupyter:

```
jupyter notebook
```

6. Create a New Notebook

Inside Jupyter: Click **New** → **Python 3** Create a notebook named *freddi_test.ipynb*

7. Test FREDDI Import

Run the following code cell:

```
import freddi
print("FREDDI imported successfully")
```

8. Example FREDDI Simulation

Example astrophysical simulation:

```
from freddi import freddi

model = freddi.Freddi(
    Mx=10,
    alpha=0.1,
    rout=1e11,
    mdot0=1e18
)

result = model.evolve()

print(result)
```

9. Plotting Results

Use matplotlib to visualize evolution:

```
import matplotlib.pyplot as plt

plt.plot(result.t, result.mdot)
plt.xlabel("Time")
plt.ylabel("Accretion Rate")
plt.title("Accretion Disk Evolution")
plt.show()
```

10. Common Problems

ModuleNotFoundError: Ensure the environment is activated. **Import errors:** Reinstall using `pip install .`

Notebook kernel mismatch: Select the correct conda environment kernel. **Permission denied:** Use `chmod +x` if required.

11. Scientific Applications

FREDDI is useful for: Accretion disk evolution X-ray binary modeling Outburst simulations Disk instability studies Time-dependent astrophysical processes

12. Important Links

Official resources: FREDDI GitHub:

<https://github.com/hombit/freddi> Jupyter Notebook:

<https://jupyter.org/> Astropy:

<https://www.astropy.org/> Matplotlib:

<https://matplotlib.org/>
